

auto_cst

5 nm peak error, **99.94%**
absorption

Autonomous NIR perfect-absorber design via a three-stage agent pipeline

Stage 1 Literature review **Stage 2** CST geometry build **Stage 3** LLM-in-loop fine-tune

Target: 1550 nm absorber · TM polarization · CST Studio Suite + OpenAI gpt-4o

The problem

Designing a NIR perfect absorber at 1550 nm

(telecom C-band, 193.41 THz) traditionally requires expert manual iteration in a full-wave EM solver — tweaking geometric parameters until the simulated reflection peak lines up with the target wavelength.

Each candidate design takes 5–15 minutes to simulate. A typical convergence path is 10–50 iterations. Across multiple shape hypotheses, this is hours-to-days of expert time per target.

PER SIMULATION

5–15 min

PER DESIGN

10–50 iters

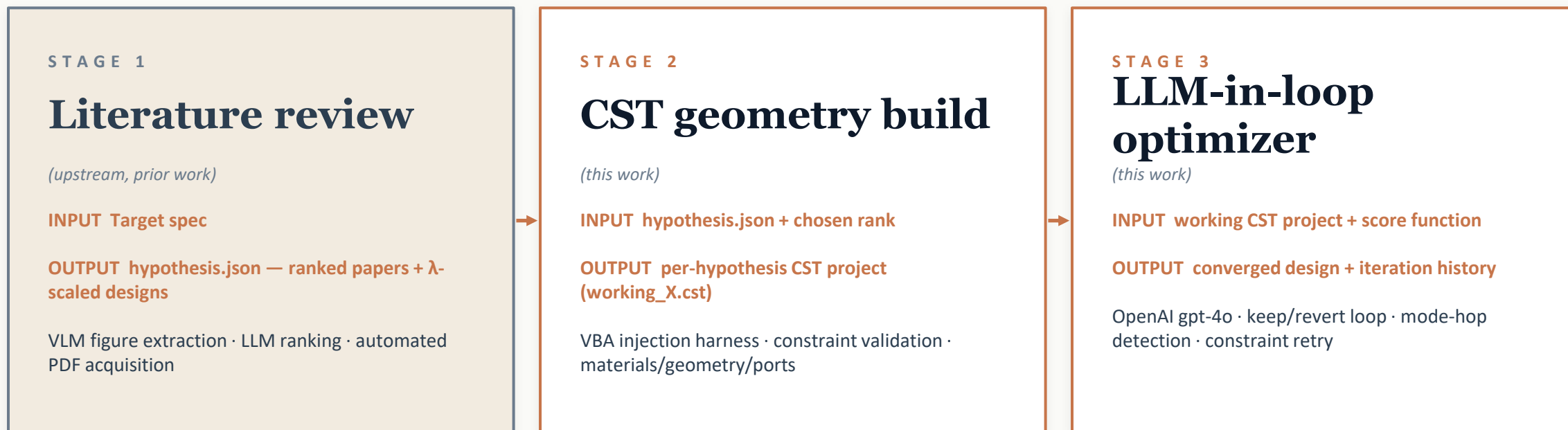
PER WAVELENGTH

3–10 designs

Goal — automate the loop end-to-end. From a one-line target spec to a converged CST project, no human in the loop.

Three-stage agent pipeline

Each stage is an autonomous LLM-driven module that hands off via a JSON contract.



This deck covers stages 2 and 3. Stage 1 output is treated as an input contract.

Stage 1 input — top-3 ranked hypotheses

Upstream lit-review agent screened 54 papers; ranked 6; handed off the top 3.

Rank	DOI	Geometry (lit review)	Score	Dimensions extracted?
1	10.1364/oe.415960	Elliptical disk supercell (4 ellipses)	0.467	Shape only — dimensions all null
2	10.1039/d2ra05617h	Metallic disk MIM on Au/SiO ₂ /Ag stack	0.442	Yes — λ -scaled by 1.32× to 1550 nm
3	10.1021/acsphotonics.8b00872	Lithography-free planar absorber	0.383	No shape, no dimensions

Insight Only rank 2 was simulation-ready. Ranks 1 and 3 had figures but no extracted dimensions — for those, stage 2 had to invent reasonable starting geometry. This made convergence harder and produced a real result we couldn't have predicted from the lit review alone.

TARGET Peak: 1550 nm (193.41 THz) FWHM: 200 nm Polarization: TM Score: $|f - target| + 0.2 \cdot \max(0, 0.90 - abs)$

Stage 2 — building the unit cell from VBA

Three hypothesis-specific runners share one MIM stack and one VBA-injection pattern.



← unit cell, period p (square lattice) →

VBA injection (one history step per item):

1. Delete default PEC box
2. StoreDoubleParameter "r", 580 (et al.)
3. Units: nm / THz / fs
4. Frequency range 100–300 THz
5. Boundary: X/Y unit cell, Z expanded open
6. Background: vacuum
7. Materials: Au_lossy, Ag_lossy, SiO₂
8. Geometry: ground brick + spacer brick + resonator
9. Floquet ports (Zmax + Zmin, 2 modes each)
10. Solver type + mesh

Three resonator shapes, one stack:

- **A** circular Ag disk (r)
- **B** rectangular Ag patch (l_x, l_y)
- **C** uniform thin Ag film (no patterning)

Solver / mesh decision rule

- Patterned absorbers (A, B): **HF Time Domain + PBA hex**
- Uniform planar (C): **HF Frequency Domain + Tetrahedral**

TD with periodic Floquet boundaries does NOT excite an absorbing mode in a fully-uniform stack.

Stage 3 — the LLM-in-loop optimizer

ChatGPT proposes parameter changes with reasoning. The runner evaluates. Score decides keep or revert.



loop until score < threshold or 5 consecutive no-improve

SCORE FORMULA

$$\text{score} = |\text{f_peak} - \text{f_target}| + 0.2 \cdot \max(0, 0.90 - \text{peak_abs})$$

First term: Hz of frequency error (dominant when peak is off-target).

Second term: penalty if peak amplitude < 0.90 (kicks in for weak resonances).

Lower is better. Convergence threshold: 0.5 THz (≈ 4 nm wavelength accuracy at 1550 nm).

PER-ITERATION DETAIL

ChatGPT input: system prompt with hypothesis-specific physics; user message with target + last 12 results + last 5 reverts

ChatGPT output: JSON {changes: {...}, reasoning: "..."}

Constraint retry: up to 2 retries with the violation message in the prompt

Subprocess CST: isolated runner.py — a CST crash never kills the agent

Disk MIM — peak landed 5 nm from target

FINAL

1554.94 nm

5 nm error (0.32 % of λ)

99.94 % peak absorption

8 iterations (4 keeps, 4 reverts) · ~14 min total wall time

FINAL DESIGN (nm)

param	baseline	converged	Δ
p (period)	993.59	1300	+306
r (disk radius)	457.05	580	+123
h (disk thickness)	105.98	105.98	0
d (SiO ₂ spacer)	112.61	100	-12.6
t_ground (Au)	100	100	0

Convergence story: iters 2–5 wasted ~6 min on a **mode hop** — the LLM proposed shrinking r to lower the frequency, but the peak *jumped 75 THz upward instead*, because a higher-order LSPR took over near the period boundary. After expanding p from 1200 to 1300 nm at iter 6 (giving the disk room), the score collapsed:

5.21 → 0.61 → 0.19 in three iterations. This case directly motivated the mode-hop detector added in retrospective improvements (slide 11).

Rect-patch MIM — landed 7 % off, learned polarization axis

PEAK

1662 nm

112 nm error (7.2 % of λ)

79.0 % peak absorption

10 iterations (5 keeps, capped without convergence)

FINAL DESIGN (nm)

p	1400
lx (initial long axis)	500
ly (now long axis)	1100
h	100
d (SiO ₂ spacer)	200
t _{ground}	100

What the LLM figured out empirically: the agent started with lx = 1100 nm as the LONG axis — but the dominant peak in the measured S_{11} responded to ly. By the end of the run the agent had *swapped the role of the two axes*, growing ly to 1100 nm and shrinking lx to 500 nm. This is genuine empirical inference — not just executing prompted physics.

Note Originally specced as elliptical disk per the rank-1 paper figure; pivoted to rect-patch after both ExtrudeCurve+Translate and Cylinder+Transform.Scale VBA paths failed (CST ActiveX 10091).

Planar Fabry-Perot — a real materials limit, not a bug

WHAT HAPPENED

- **Time Domain solver:** absorptance identically zero across 100–300 THz ($|S_{11}|^2 = 1.0$). TD with periodic Floquet boundaries on a fully-uniform structure does not excite an absorbing mode.
- **Frequency Domain solver:** tiny absorption (< 5 %) pegged to spectrum edge. Real signal but no resonant peak.
- **Agent loop:** 8 attempts varying $d \in \{267, 300, 450, 600 \text{ nm}\}$ and $t_{\text{top}} \in \{8, 12 \text{ nm}\}$. Score plateaued at 106.76 (the noise-floor score) for all of them.

WHY (the physics)

The lithography-free planar absorber relies on *impedance matching* between a thin top metal film and free space at cavity resonance.

At 1550 nm, Au has $\epsilon \approx -95 + 11i$ (Johnson & Christy fit) — strongly Drude-dispersive.

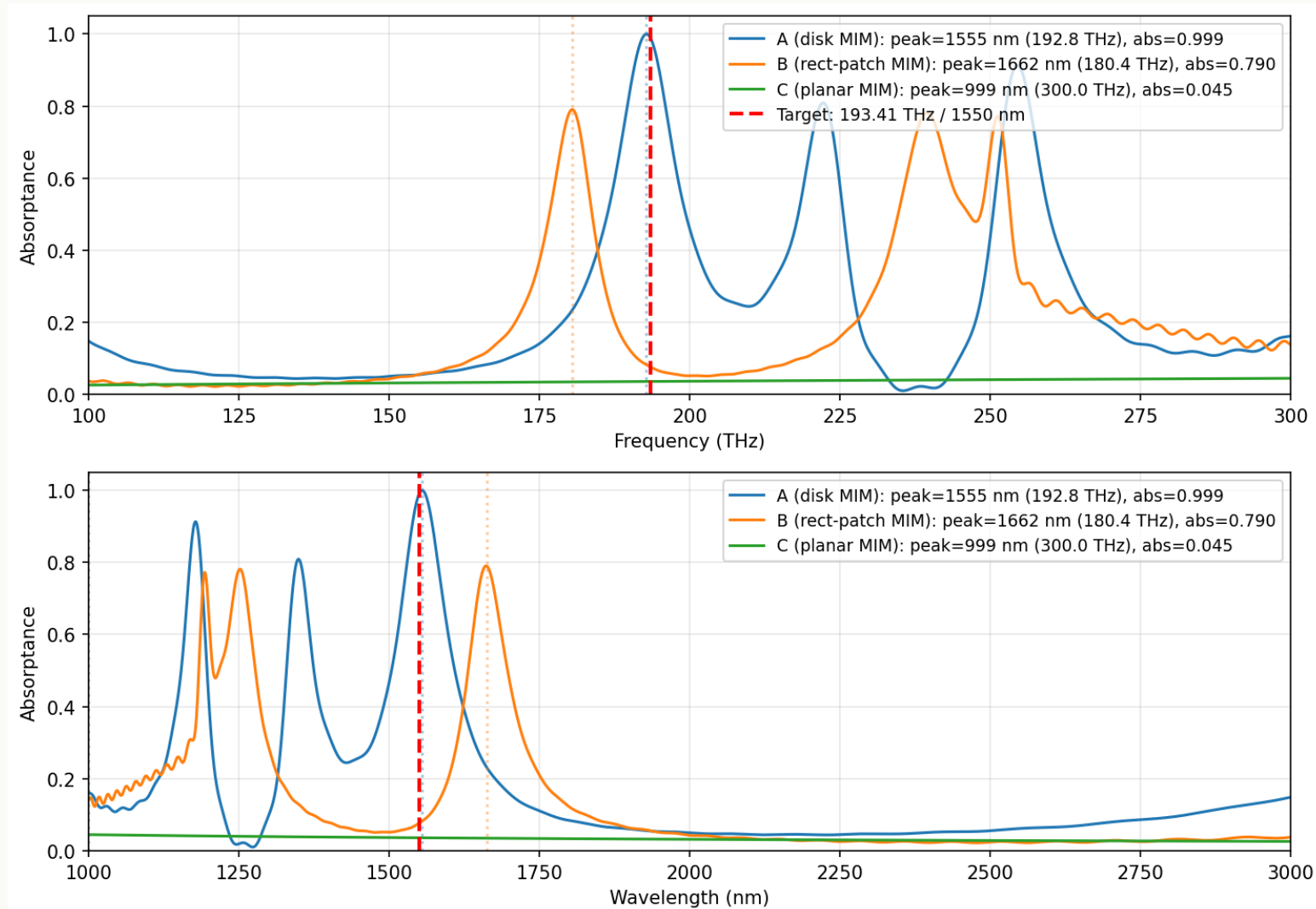
Our materials use a constant- σ model (DC values + CST's high-frequency surface-impedance approximation). This is fine for **patterned LSPR-driven absorbers** (A and B work because the LC mode is dominant) — but for the impedance-matching cavity, the ratio of real to imaginary ϵ is wrong by an order of magnitude.

Result: high reflection across the band. A real physical/materials limitation, not a code bug.

The unblock: load CST's built-in dispersive Au / Ag from the material library. `nir/probe_drude_vba.py` automatically tries 9 candidate VBA syntaxes — likely takes one minute to find the right one in any given CST install.

All three designs — final spectra side by side

Vertical red dashed = target 1550 nm / 193.41 THz · vertical dotted = each design's measured peak.



Building robust CST automation

Five failure modes hit during the run, and the defensive code that now catches each one.

01

cst.results.ProjectFile silently breaks with relative paths

Result lookup raises “does not exist for run id=N” even though the data is in the file.

FIX Wrapper open_results() always passes Path(...).resolve()

02

Parameter injection invalidates run_id=0; API default is run_id=0

Real data lives at run_id=1+. Default get_result_item silently returns nothing.

FIX Wrapper queries get_all_run_ids(), requests highest first

03

Solver/mesh combo matters — wrong combo gives zero absorption with no error

TD+PBA hex for patterned absorbers; FD+Tetrahedral for uniform planar.

FIX Hypothesis-specific solver/mesh dispatch in runner.py

04

Constant- σ Au/Ag wrong for impedance-matching designs

OK for plasmonic-resonance designs (A, B). Catastrophic for thin-film cavity (C).

FIX probe_drude_vba.py auto-tries 9 dispersive-material syntaxes

05

LLM extrapolates trends across mode hops

Iters 2–5 of hypothesis A wasted on a higher-order LSPR masquerading as the fundamental.

FIX Mode-hop detector auto-warns the LLM in the user message

CONCLUSIONS

5 nm peak error in ~14 minutes

of CST + LLM time, fully autonomous from a one-line target spec.

Hypothesis matters

Patterned absorbers worked (A converged, B partial). The lithography-free planar fell out — the materials model couldn't capture impedance matching. A genuinely different problem class.

LLM can do empirical inference

Hypothesis B's agent swapped the roles of I_x and I_y mid-run after seeing how the spectrum responded — not just executing prompted physics.

Defensive wrappers pay off

Every silent CST footgun (relative-path indexing, `run_id=0`, solver/mesh combos) cost real wall-time the first time. Once wrapped, they're free for everyone after.

FUTURE WORK

- Run `probe_drude_vba.py` to fix dispersive Au/Ag and unblock hypothesis C
- Add multi-resonator topology (Salisbury overlay or dual-disk supercell) to broaden the 105 nm FWHM toward the 200 nm target
- Generalize to UV / mid-IR / GHz bands with per-band solver/mesh recipes